732A96/TDDE15 ADVANCED MACHINE LEARNING

EXAM 2026-10-27

Teacher

Jose M. Peña. He will visit the room and answer questions on the phone.

Grades

- For 732A96 (A-E means pass):
  - A=19-20 points
  - B=17-18 points
  - C=14-16 points
  - D=12-13 points
  - E=10-11 points
  - F=0-9 points
- For TDDE15 (3-5 means pass):
  - 5=18-20 points
  - 4=14-17 points
  - 3=10-13 points
  - U=0-9 points

In each question, full points requires clear and well motivated answers and commented code.

Instructions

- This is an individual exam. No help from others is allowed. No communication with others is allowed. Answers to the exam questions may be checked for plagiarism.
- This is an anonymous exam. Do not write your name on it.
- The answers to the exam should be submitted in a single PDF file. You can make a PDF from LibreOffice (similar to Microsoft Word). You can also use Markdown from RStudio (no support is provided though). Include important code needed to grade the exam (inline or at the end of the PDF file).

Allowed help

Everything on the course web page. Your individual and group solutions to the labs. This help is available on the corresponding directories of the exam system.

1. Probabilistic Graphical Models (5 p)

Consider the Bayesian network with binary variables and structure

$$S \to L, \qquad T \to E \leftarrow L, \qquad E \to X.$$

and conditional probability tables

Factor	Values
$P(S)$	$P(S = 1) = 0.5$
$P(L \mid S)$	$P(L = 1 \mid S = 1) = 0.2, \quad P(L = 1 \mid S = 0) = 0.05$
$P(T)$	$P(T = 1) = 0.1$
$P(E \mid T, L)$	$P(E = 1 \mid T, L) = 1$ if $T = 1$ or $L = 1$, otherwise 0
$P(X \mid E)$	$P(X = 1 \mid E = 1) = 0.9, \quad P(X = 1 \mid E = 0) = 0.1$

Compute $P(L \mid X = 1)$ using the variable elimination algorithm seen in the lectures. Eliminate the variables in the order S, T, E. For each step write (i) the factors multiplied, (ii) the scope of the product, (iii) the new factor as a table.

2. Hidden Markov Models (5 p)

Consider a hidden Markov model for the weather. The hidden state $Z_t \in \{R, S\}$ is rain (R) or sun (S), and the observation $X_t \in \{n, y\}$ is whether a person carries an umbrella (y) or not (n), for $t = 1, 2, 3$.

Quantity	Values
Initial distribution of Z_1	$P(Z_1 = R) = 0.5$, $P(Z_1 = S) = 0.5$
Transition $P(Z_t \mid Z_{t-1})$	$R \to R$: 0.7, $R \to S$: 0.3
	$S \to R$: 0.4, $S \to S$: 0.6
Emission $P(X_t = y \mid Z_t)$	R: 0.9, S: 0.2

Use the forward-backward algorithm seen in the lectures to compute the filtered distribution of the weather at each time step. The observed sequence is $X_1 = y$, $X_2 = y$, $X_3 = n$.

3. Reinforcement Learning (5 p)

This exercise is about the Monte Carlo method (a.k.a. MC control) seen in the lectures. The file `MCcontrol_MLAS2025_Scrambled.R` contains an implementation of the algorithm. However, the code lines for the functions `MC_episode` and `MC_control` are scrambled. Your task is to put them in the right order, and then run the algorithm in the environment A seen in the labs.

4. Gaussian Processes (5 p)

The derivative or gradient of a function g at a point x is defined as the limit of

$$\frac{g(x+h)-g(x)}{h}$$

when h goes to 0. It can be approximated by

$$\frac{g(x+\epsilon)-g(x-\epsilon)}{2\epsilon}$$

when ϵ is small. Use this approximation to implement gradient descent to select the bandwidth l of the squared exponential kernel of a Gaussian process for classification. Specifically, define

$$l = \exp(K)$$

and optimize K.

Use the `kernlab` package and the iris dataset. Use only two classes from the iris dataset. Use the negative log-likelihood as loss function: For $y_i \in \{0,1\}$ and predicted probability $p_i = P(y_i = 1 \mid x_i)$, define the loss as

$$L = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right].$$

Your code should include three functions: Loss calculation, gradient calculation, and parameter updating. You are free to make any choices that are not specified above. You do not need to run your code, but make sure that it includes appropriate comments for readability.